

Ahmad Al Najjar

Computer Science Student

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EDUCATION

UNIVERSITY OF UTAH

Salt Lake City, UT

B.S in Computer Science & Minor in Financial Technology (FinTech)

Notable Computer Science Courses

- Software Developing II
- Data Structures and Algorithms
- Image Processing

EXPERIENCE

- **UCUBESAT CLUB** **2023-Present**
University of Utah
 - Developed flight software using NASA's Embedded flight language F' (f-Prime).
 - Collaborated directly with NASA's F' (f-Prime) development team, contributing to debugging and optimizing mission-critical software.
 - Applied insights from NASA JPL's F' workshop to improve software reliability, ensuring adherence to industry standards.
- **TEACHING ASSISTANT - FINANCIAL TECHNOLOGY COURSE** **2023-Present**
University of Utah
 - Provided support for the instructor in preparing course materials.
 - Proactively engaged and instructed in industry-leading FinTech Topics.

PROJECTS

- **SPRITE EDITOR - C++/ GNU & G++**
 - Engineered a sprite editor using C++ with Qt for cross-platform GUI development.
 - Applied object-oriented programming principles to create a modular and maintainable codebase.
 - Developed a custom rendering pipeline for real-time pixel manipulation and layering with 50% reduction in rendering latency.
 - Implemented an event-driven architecture to handle user interactions efficiently.
- **SOCCER ANALYSIS - PYTHON**
 - Developed soccer footage analysis tool using YOLO, achieving 85% accuracy in object detection.
 - Developed an object detection model using YOLO to detect players, referees, and the ball, achieving high accuracy and minimizing false positives.
 - Implemented KMeans clustering to segment and classify players and accurately assign them to teams with a 90% success rate.
 - Calculated team ball possession percentages, providing insightful metrics for game analysis.
- **SPREADSHEET APPLICATION - C#**
 - Developed a spreadsheet application in C# using MAUI and .NET 8 framework
 - Implemented MVC architecture, increasing code maintainability by 40%.
 - Demonstrated proficiency in backend and frontend separation.

TECHNICAL SKILLS

- **LANGUAGES:** C++, Java, C#, Python, HTML/CSS, SQL, MySQL,
- **LIBRARIES/Frameworks:** PyTorch, TensorFlow, NumPy, KMeans, YOLO, JUnit, F' (f-Prime), .NET
- **DEV TOOLS:** Git, G++ Docker, Visual Studio, VS Code, IntelliJ, AWS, Microsoft Products, Embedded
- **CERTIFICATIONS:** F-prime Flight FrameWork WorkShop, NASA JPL